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Globally time-reversible fluid simulations with smoothed particle hydrodynamics ☆

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ABSTRACT

This paper describes an energy-preserving and globally time-reversible code for weakly compressible smoothed particle hydrodynamics (SPH). We do not add any additional dynamics to the Monaghan's original SPH scheme at the level of ordinary differential equation, but we show how to discretize the equations by using a corrected expression for density and by invoking a symplectic integrator. Moreover, to achieve the global-in-time reversibility, we have to correct the initial state, implement a conservative fluid-wall interaction, and use the fixed-point arithmetic. Although the numerical scheme is reversible globally in time (solvable backwards in time while recovering the initial conditions), we observe thermalization of the particle velocities and growth of the Boltzmann entropy. In other words, when we do not see all the possible details, as in the Boltzmann entropy, which depends only on the one-particle distribution function, we observe the emergence of the second law of thermodynamics (irreversible behavior) from purely reversible dynamics.

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1. Introduction

The original smoothed particle hydrodynamics (SPH) scheme, now called weakly compressible SPH (WCSPH), was developed in 1970s by R. Gingold and J. Monaghan [1]. The scheme used the

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leap-frog time integration and relied on an artificial term stabilizing the fluid in the presence of shocks. Many other stabilization strategies were conceived. In the δ-SPH method [2], diffusion terms are added to the continuity equation, and methods based on Riemann solvers [3]. These approaches often provide results in reasonable agreement with experimental data or other numerical methods. Unfortunately, they often come at the cost of increased complexity and necessity to fine-tune additional numerical parameters [4].[1] Failure in such fine-tuning can result in unrealistic dissipation, or large energy oscillations unless a convenient limiter is used [7].

Another problem of SPH (especially in solid mechanics) is the so called tensile instability, where simulations are unable to withstand a dilation of volume (leading to clamping of particles inside the solid body) [8]. A solution is to use the Lagrangian kernels, which depend on the Lagrangian distances between the particles instead of the Eulerian distances [9][10]. The price to pay is that the Lagrangian kernels become too distorted for large deformations, which can be avoided by switching between the Lagrangian and Eulerian iterations [11]. In this manuscript, we do not consider solid mechanics, which is why we can work with the standard Eulerian kernels.

Novelty of this manuscript lies mainly (to our best knowledge) in the global reversibility of the SPH scheme and in the thermodynamic interpretation of the results (growth of Boltzmann entropy despite the reversibility). To achieve the global reversibility, we had to formulate SPH as a symplectic system, to correct the way density is typically treated, to use the fixed-point arithmetic, and to implement a conservative wall-fluid interaction. Each of these steps is necessary for the global reversibility. The second main new point lies in the thermodynamic interpretation of the emergence of irreversibility that we observe. Although the underlying dynamics is reversible and the Liouville entropy (that takes into account positions of all particles) does not grow, the Boltzmann entropy (that no longer takes into account the individual particles) grows. This emergence of the second law of thermodynamics is caused by the information loss between the complete Liouville description and the averaged Boltzmann description of the particle system.

Although the reversible (non-dissipative) terms in the SPH simulations typically do not alter the total energy appreciably, they can cause a sort of irreversibility. WCSPH simulations contain the evolution equation for the discrete density and can also contain an evolution equation for particle energy, but if the underlying numerical integrator is not symplectic, it changes the phase volume, which causes irreversible evolution of the distribution function [12,13]. In this manuscript, we construct a globally reversible symplectic SPH scheme, which can be used also for backward simulations and leads to the initial state of the particles, and thus it does not introduce numerical dissipation in the above sense.

The symplecticity can be achieved by a particular way to calculate the pressure. There are essentially two different ways, how pressure can be calculated in WCSPH [14]. A commonly used approach is to update the density iteratively, using the velocities obtained from the discretized balance of momentum. Alternatively, one can compute density directly as an SPH interpolation of particle masses [15]. The latter approach is less common, but it leads to a symplectic structure of the system of SPH differential equations, which makes the WCSPH discretization completely stable even in the absence of viscosity or stabilization. Although the former method leads also to a Hamiltonian system, it is not symplectic, but only Poisson, which makes it harder to construct a structure-preserving geometrical numerical integrator [12]. Another a bit more complicated symplectic scheme was proposed in [16], where the density is coupled with another field (similarly as position is coupled with momentum).

However, standard symplectic WCSPH shows unphysical behavior at free surfaces, where constant functions are not correctly interpolated [14]. We suggest two ways how to correct this issue. The first method modifies the density by an appropriate integration constant, which is easy to implement. The second, and more complex method, renormalizes the initial state by solving a certain nonlinear problem using Newton's method. We also demonstrate that a careful implementation of the symplectic WCSPH scheme, using the corrected treatment of free surfaces and the fixed-point (as opposed to the more usual floating-point) arithmetic leads to global time-reversibility. This can be considered as an additional symmetry-preserving feature alongside the conservation of energy, mass, momentum, and angular momentum. The purpose of this paper is to show key advantages of the symplectic WCSPH compared to the standard approach.

The symplectic WCSPH with a proper treatment of free boundaries, conservative wall-fluid interaction, and fixed-point arithmetic becomes globally reversible in time. For instance, the breaking-dam simulation can be reversed to recover the initial conditions. Although the resulting code is energy-preserving and reversible, it can still be considered dissipative in the sense that the Boltzmann entropy of the SPH particles grows. In other words, we observe the emergence of the second law of thermodynamics from purely reversible dynamics. In the case of the breaking dam simulation, the particles eventually occupy the bed of the container and their velocity distribution is approximately Maxwellian. We provide a reasoning for this apparent paradox between reversibility and irreversibility by means of information theory at the end of the manuscript.

The energy budget in SPH simulations is of great interest because it indicates which degrees of freedom lose energy and which gain it [17]. Dissipative terms (like the viscous terms or the diffusive terms) typically reduce the total energy, but sometimes the kinetic energy decreases and compressional energy increases while in other cases the kinetic energy increases. Although it may be tempting to link the decrease of a particular kind of energy (for instance kinetic energy) with dissipation, such link can not hold in general, since sometimes the particular kind of energy increases and sometimes decreases. The equilibrium partitioning of the total energy budget can be actually estimated from the virial theorem [18]. Therefore, the initial conditions determine which part of the energy will grow and which will drop. For isolated systems, it is the growth of entropy that indicated dissipation while in isothermal systems the total free energy decreases [19]. In this manuscript, we discuss how entropy grows even in reversible isolated systems.

We demonstrate the various versions of WCSPH on dam-break and Gresho vortex benchmarks. All SPH codes used to make this paper is freely available within the new SmoothedParticles.jl package [20] written in the Julia programming language [21]. But let us first recall the WCSPH equations in the following Section.

2. WCSPH for inviscid fluids

The standard weakly compressible spatial semi-discretization of Euler fluid equations in a uniform gravitational field is [22]:

$$\partial_t \rho_a = \sum_b m_b (\boldsymbol{u}_a - \boldsymbol{u}_b) w'(r_{ab}) \boldsymbol{e}_{ab}, \tag{1a}$$

[1] It is possible that the intimate connection between the parameters in the diffusive and viscous term might be connected with the alternatives to the Navier-Stokes equations where Laplacian-like terms are present in both the momentum equation and the equation of continuity [5,6].

$$\partial_t \boldsymbol{u}_a = \sum_b m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2} \right) w'(r_{ab}) \boldsymbol{e}_{ab} - g\hat{\boldsymbol{z}}, \tag{1b}$$

$$\partial_t \boldsymbol{r}_a = \boldsymbol{u}_a, \tag{1c}$$

where the pressure is a function of density via the barotropic formula,

$$p(\rho) = \frac{c^2 \rho_0}{7} \left[\left(\frac{\rho}{\rho_0} \right)^7 - 1 \right]. \tag{2}$$

Constant ρ_0 stands for the referential density, g is the gravitational acceleration, and c is the numerical speed of sound (typically chosen as ten times the characteristic flow speed U_{char}). It is well known that the system of ordinary differential equations (1) conserves the total energy in the form

$$\mathcal{H} = \sum_a m_a \left(\frac{v_a^2}{2} + \epsilon_a + gz \right), \tag{3}$$

where

$$\epsilon_a = \frac{c^2}{42} \left[\left(\frac{\rho}{\rho_0} \right)^6 + \frac{6\rho_0}{\rho} \right]$$

is the potential of internal energy (also called compressional) satisfying $\mathrm{d}\epsilon_a = \frac{p_a}{\rho_a^2} \mathrm{d}\rho_a$. Moreover, for $g = 0$, the momentum

$$\boldsymbol{\mathcal{M}} = \sum_a m_a \boldsymbol{u}_a,$$

and angular momentum

$$\boldsymbol{\mathcal{L}} = \sum_a m_a \boldsymbol{r}_a \times \boldsymbol{u}_a$$

are conserved as well. Naturally, gravitational force can deliver some momentum to the fluid, while the equal opposite reaction of the fluid on Earth is considered grounded.

The WCSPH equations (1) represent a Hamiltonian system, which can be checked for instance using program [23]. But it is not a symplectic system because the mass density is treated as a state variable. If, on the other hand, the density would always be calculated from the current positions of the particles, then only the positions and momenta would be state variables and the system of equations would be symplectic. Symplecticity would make it easier to choose a geometric structure-preserving numerical integrator, for instance using the Störmer-Verlet scheme [24,12].

A symplectic form of WCSPH is obtained by solving the ordinary differential equation for density (1a),

$$\rho_a = \sum_b m_b w(r_{ab}), \tag{4}$$

instead of updating ρ_a iteratively by (1a). However, this closed expression for density (4) is rarely used in practice because it leads to questionable behavior near free surfaces [14]. This happens because free boundary particles are under-occupied and, according to (4), they have $\rho \ll \rho_0$. Therefore, they are equipped with large internal energy and, affected by its negative gradient, they start to vibrate. A remedy is provided in the following section, which leads to the possibility to use the symplectic form of WCSPH.

3. Treatment of free surfaces

3.1. Solving the equation for ρ_a with an appropriate integration constant

The issue of unphysical behavior at free surfaces can be corrected by choosing an appropriate integration constant when solving the differential equation for ρ_a,

$$\rho_a = \sum_b m_b w(r_{ab}) + C_a, \tag{5}$$

where

$$C_a = \rho_0 - \rho_a|_{t=0}.$$

Indeed, since WCSPH based on (1a) typically assumes zero internal forces at $t = 0$, it is actually (5) and not (4) that is equivalent to the standard WCSPH formulation, since (4) leads to non-zero density gradients near the boundary. Although schemes based on (1a) and (5) are completely equivalent at the ODE level, they become very much different once time discretization is considered. It appears that (1a) is currently used by most researchers, despite the advantages of Equation (5):

1. Inferring ρ from positions using Equation (5) does not accumulate the density error.
2. The system of SPH evolution equations becomes symplectic (and not only Hamiltonian).
3. Therefore, symplectic structure-preserving integrators can be employed, which leads to simulations globally reversible in time.

However, even if we use the correct closed formula for density (5) and use a symplectic integrator, are facing another obstacle. For problems involving free surfaces, integration constant C_a is typically much bigger for boundary particles and, therefore, whenever such particle submerges into interior of the fluid body, it generates a small void pocket around itself. This unwanted numerical artefact is removed in the following section.

3.2. Initial state correction (ISC)

The numerical artefact that particles that are originally at the boundary are treated differently than the rest of the particles, even if the submerge into the in the interior of the fluid, is rooted in the difference in the C_a constants at the beginning. Therefore, we shall remove that artifact by correcting the initial state.

A solution is to correct the initial positions by a vector field $\delta\boldsymbol{x}$ such that

$$\rho_0 = \sum_b m_b w \left(\left| \boldsymbol{x}_a + \delta\boldsymbol{x}_a - \boldsymbol{x}_b \right| \right).$$

These non-linear equations can be solved by a strategy based on the Newton's method, details of which follow. Linearization with respect to $\delta\boldsymbol{x}_a$ leads to

$$\rho_0 - \rho_a \doteq \sum_b m_b (\delta\boldsymbol{x}_a - \delta\boldsymbol{x}_b) \cdot w'(r_{ab}) \boldsymbol{e}_{ab}.$$

On the right hand side, we identify the discrete divergence operator $\tilde{D}^1$

$$-\rho_a \tilde{D}_a^1 \delta\boldsymbol{x} \doteq \rho_0 - \rho_a, \tag{6a}$$

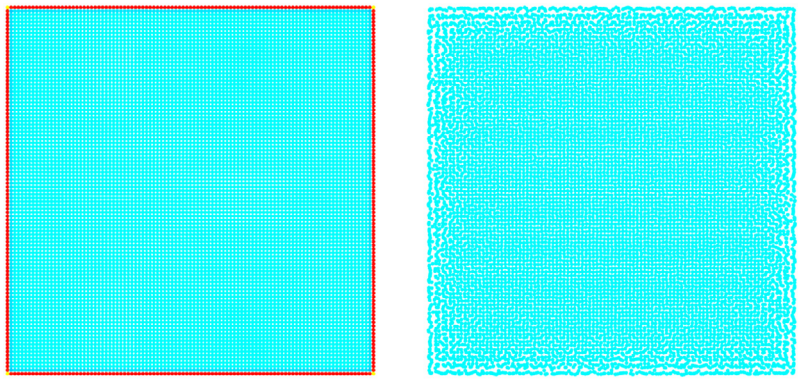
see [14] for details. This linear problem for unknowns $\delta\boldsymbol{x}$ is underdetermined, so shall be restricted to $\delta\boldsymbol{x}$ in the form of a discrete gradient

$$\delta\boldsymbol{x}_a = \sum_b m_b \left(\frac{\phi_a}{\rho_a^2} + \frac{\phi_b}{\rho_b^2} \right) w'(r_{ab}) \boldsymbol{e}_{ab},$$

which can be also rewritten as

$$\delta\boldsymbol{x}_a + \frac{1}{\rho_a} G_a^1 \phi = 0. \tag{6b}$$

Sparse linear system (6) for $\delta\boldsymbol{x}$ and ϕ has to be solved iteratively,



(a) before ISC (b) after ISC

Fig. 1. Applying ISC to particles in square arranged regularly. Density (4) is shown in color (red to blue). Initially, it varies due to boundary effect but becomes everywhere constant after 7 iterations of ISC (up to relative error less than 10^{-10}). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

updating $\boldsymbol{x}$ and reevaluating ρ according to (4), until $\rho \doteq \rho_0$ with satisfactory precision. This initial state correction (ISC) procedure has been tested in our numerical examples, and the procedure converges quadratically except for problems with too much symmetry (for instance rectangle with particles arranged in square grid nearly always diverges).[2] The whole procedure may be then summarized as follows (see Fig. 1):

Initial state correction (ISC)

1. set $|\delta \boldsymbol{x}_a| < \frac{\delta r}{10}$ randomly for all a
2. repeat until sufficient precision is acquired:
3. $\quad \boldsymbol{x}_a = \boldsymbol{x}_a + \delta \boldsymbol{x}_a$ for all a
4. $\quad$ recompute neighbor list
5. $\quad$ find $\rho_a = \sum_b m_b w(r_{ab})$ for all a
6. $\quad$ solve $N(d+1) \times N(d+1)$ linear problem

$$\begin{bmatrix} I & \frac{1}{\rho} G^1 \\ -\rho \tilde{D}^1 & 0 \end{bmatrix} \begin{bmatrix} \delta \boldsymbol{x} \\ \phi \end{bmatrix} = \begin{bmatrix} 0 \\ \rho_0 - \rho \end{bmatrix}$$

This computation is expensive, but is performed only once per simulation. In place of the linear system with 2×2 block matrix, one could also solve directly for ϕ:

$$-\rho \tilde{D}^1 \frac{1}{\rho} G^1 \phi = \rho_0 - \rho.$$

However, the product $A = -\rho \tilde{D}^1 \frac{1}{\rho} G^1$ leads to a matrix which is much more dense. A matrix-free iterative method could be used without evaluating the product directly, is a future possibility.

Let us note that after ISC, all constant fields will be correctly interpolated, at least initially. Weak compressibility helps to preserve this trait approximately in a way that does not accumulate error provided (4) is used. In this sense, ISC can be considered an alternative to (zeroth order) operator renormalization with the benefit that it does not break the symmetry of smoothing kernel. Unfortunately, this technique has two drawbacks. Firstly, it may slightly deform geometry in an unwanted way. Secondly, adapting it to a concrete setting may be difficult — for instance, when an anti-clump term is included.

[2] For these cases, we recommend to initially disrupt particle positions by a random noise.

Both symplecticity of the system of equations and ISC improve the numerical solutions, but in order to obtain a globally reversible behavior we have to tackle a further problem caused at the walls of the simulation box, as in the following section.

3.3. Wall modelling

In WCSPH, walls can be modeled by a layer of dummy particles which behave as if they belonged to the fluid, except that their positions and velocities are fixed. In this approach, it is hoped that the forces preventing compression will deflect any incoming particle. This method, of course, violates balance of momentum, but this is plausible, since we can imagine that any momentum yielded by the fluid is grounded. More importantly though, dummy particles violate conservation of energy. To see this, consider a wave stopped by a wall. The fluid acts on the solid with equal and opposite force. Therefore, a wall particle a is subject to acceleration, which should change its velocity after time step δt to

$$\boldsymbol{v}_a = \delta t \frac{\boldsymbol{F}_a}{m_a}.$$

To preserve wall integrity, this velocity must be annulled, reducing the total energy by a small kinetic contribution. From this, we see that dummy particles are essentially dissipative.

One could argue that this is still physically reasonable because every fluid-solid collision creates a sound wave, leading to loss of energy. However, it is dubious whether such effect is correctly modeled by dummy particles. Moreover, it complicates efforts to make a strictly energy-conservative scheme.

An alternative approach uses a fluid-solid force inspired by the Lennard-Jones potential

$$\boldsymbol{F}_{a \to b} = -\frac{E_{\text{wall}}}{r_{ab}^2 + \varepsilon^2} (s_{ab}^2 - s_{ab}) \boldsymbol{r}_{ab}$$

by which a wall particle a acts onto a fluid particle b. Here,

$$s_{ab} = \begin{cases} \frac{r_{\text{wall}}^2 + \varepsilon^2}{r_{ab}^2 + \varepsilon^2} & \text{if } r_{ab} < r_{\text{wall}}, \\ 1 & \text{otherwise.} \end{cases}$$

r_{ab} is the distance between a wall particle and a fluid particle, and $E_{\text{wall}}, r_{\text{wall}}$, and ε are numerical parameters [25].[3] Except for this

[3] In our dam-break simulation, we choose $E_{\text{wall}} = 10gh$ (g being the gravitational acceleration and h the height of the water column), $r_{\text{wall}} = 0.95dr$ (dr being the typical distance between particles), and $\varepsilon = 10^{-6}$.

interaction, wall particles are neither considered in the balance of mass (1b) nor in the calculation of density (5). Unlike the dummy particle approach, this solid wall modeling is conservative, with a modified energy

$$\mathcal{H} = \sum_{a\in\text{fluid}} m_a \left(\frac{v_a^2}{2} + \epsilon_a + gz\right) + \sum_{a\in\text{wall}} \sum_{b\in\text{fluid}} E_{\text{wall}} \left(\frac{1}{4}s_{ab}^2 - \frac{1}{2}s_{ab} + \frac{1}{4}\right), \tag{7}$$

and the wall-particle force is just minus the derivative of the wall-particle interaction energy. To see this, compute the total differential

$$\begin{aligned}
\mathrm{d}\mathcal{H}_{\text{wall}} &= \sum_{a\in\text{wall}} \sum_{b\in\text{fluid}} E_{\text{wall}}\, \mathrm{d}\left(\frac{1}{4}s_{ab}^2 - \frac{1}{2}s_{ab} + \frac{1}{4}\right) \\
&= \sum_{a\in\text{wall}} \sum_{b\in\text{fluid}} E_{\text{wall}} \left(\frac{s_{ab}}{2} - \frac{1}{2}\right) \mathrm{d}s_{ab} \\
&= -\sum_{a\in\text{wall}} \sum_{b\in\text{fluid}} E_{\text{wall}} \left(\frac{s_{ab}}{2} - \frac{1}{2}\right) \frac{r_{\text{wall}}^2 + \epsilon^2}{\left(r_{ab}^2 + \epsilon^2\right)^2} \mathrm{d}r_{ab}^2 \\
&= -\sum_{a\in\text{wall}} \sum_{b\in\text{fluid}} E_{\text{wall}} \left(s_{ab}^2 - s_{ab}\right) \frac{1}{r_{ab}^2 + \epsilon^2} \boldsymbol{r}_{ab} \cdot \mathrm{d}\boldsymbol{r}_{ab}
\end{aligned}$$

and hence, we have, for every fluid particle b:

$$-\frac{\partial \mathcal{H}_{\text{wall}}}{\partial \boldsymbol{x}_b} = -\sum_{a\in\text{wall}} E_{\text{wall}} \left(s_{ab}^2 - s_{ab}\right) \frac{1}{r_{ab}^2 + \epsilon^2} \boldsymbol{r}_{ab} = \sum_{a\in\text{wall}} \boldsymbol{F}_{a\to b}$$

This makes it potentially advantageous for problems involving inviscid flows.

Note that differential equations (1) (with or without the explicit density integration (4)) automatically conserve the Hamiltonian (or energy), due to their Hamiltonianity (in particular symplecticity) [14]. Indeed, the energy is conserved due to the skew-symmetry of the underlying Poisson bracket, and the momentum is conserved if the Hamiltonian is invariant with respect to spatial translations [12,26]. Or course, the conservativeness can be also checked manually by directly evaluating the conservation laws.

The values of the wall-interaction parameters depend on the particular situation. The wall energy must be much higher than typical energy of a particle colliding with the wall so that particles effectively bounce off the wall. The r_{wall} parameter must be, on the other hand, smaller than the typical distance between the particles so that the wall-particle interactions do not disturb the particle-particle interactions.

In our case, when we seek a globally reversible-in-time simulation, the conservativeness of the fluid-wall interaction becomes important.[4] However, even if we use the correct close formula for density (1a), the initial state correction, and the conservative fluid-wall interaction, the symplectic Störmer-Verlet scheme is not reversible globally in time because of the errors caused by the floating-point arithmetic. The following section contains the final ingredient necessary for that reversibility, the symplectic Störmer-Verlet integrator with fixed-point arithmetic.

4. Symplectic integrator and fixed-point arithmetic

The symplectic character of evolution equations (1b) and (1c) is preserved in symplectic numerical schemes, for instance the classical Verlet scheme [27]. Consequences are for instance the long-time stability of the trajectories and conservation of integrals of motion (energy, momentum, and angular momentum, among others) [12]. However, even application of the Verlet scheme in SPH does not lead to globally in-time reversible simulations due to the floating-point errors, but a remedy is in the fixed-point arithmetic, as we show below.

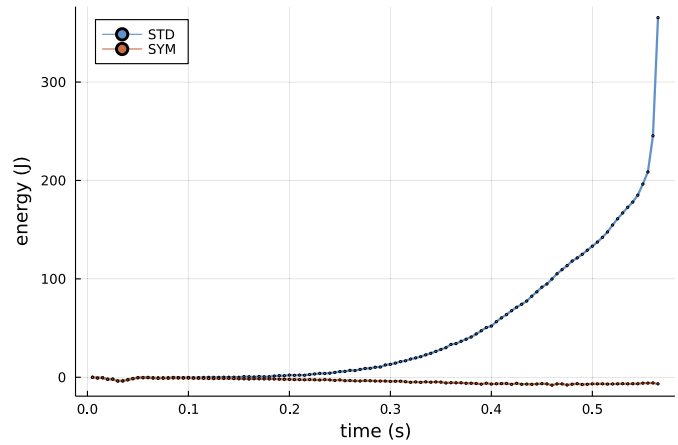


Fig. 2. Energy growth in standard WCSPH (STD) and symplectic WCSPH (SYM) in dambreak simulation with zero viscosity and parameters from Table 1. Without a stabilization of some sort, STD diverges, whereas SYM is stable.

The advantage of the direct update of density to the integration of the continuity equation is that the former way leads to a symplectic scheme while the latter to a non-symplectic (albeit still Hamiltonian) scheme. Although the scheme with integration of the continuity equation is still Hamiltonian, the underlying Poisson bracket is non-canonical (Poisson geometry), and it is more difficult to construct numerical schemes respecting the geometrical structure [12]. For instance, the standard symplectic schemes like Störmer-Verlet do not preserve the Poisson structure, but they preserve the symplectic structure.

The usual discretization of the SPH equations using the Störmer-Verlet scheme is

1. $\boldsymbol{u}(t_{m+\frac{1}{2}}) = \boldsymbol{u}(t_m) + \frac{1}{2}\delta t\, \boldsymbol{a}(\boldsymbol{r}(t_m))$
2. $\boldsymbol{r}(t_{m+1}) = \boldsymbol{r}(t_m) + \delta t\, \boldsymbol{u}(t_{m+\frac{1}{2}})$
3. $\boldsymbol{u}(t_{m+1}) = \boldsymbol{u}(t_{m+\frac{1}{2}}) + \frac{1}{2}\delta t\, \boldsymbol{a}(\boldsymbol{r}(t_{m+1}))$,

where $\boldsymbol{a}$ is the acceleration composed of internal, gravitational, and wall forces. The scheme is of second order and symplectic, and therefore it conserves energy $\mathcal{H}_{\delta t}$ with error

$$\mathcal{H}_{\delta t}(\boldsymbol{r}, \boldsymbol{u}) - \mathcal{H}(\boldsymbol{r}, \boldsymbol{u}) = O\left(\delta t^2\right)$$

that does not depend explicitly on time, provided that δt is sufficiently small [28].[5] If we took ρ as a separate state variable with its own evolution equation (1a) (usual WCSPH), symplecticity of the system of equations would be violated. The usual WCSPH equations are still Hamiltonian (generated by a Poisson bracket and a Hamiltonian), as checked by program [23], but they are not symplectic and they thus the (symplectic) Verlet scheme does not preserve their geometric structure, which causes energy to grow exponentially, unless some stabilization is used - see Fig. 2.[6]

[4] Note that the results about global time-reversibility would stay the same if we chose a system without boundary, where no wall-fluid treatment would be necessary.

[5] Here, we rely on the fact that $p = p(\rho) = p(\rho(\boldsymbol{r}))$ using the closed expression (5).

[6] In [22], it is suggested to correct this error by a mid-step extrapolation of density. However, it should be noted this approach adds additional inaccuracy to the scheme.

Table 1
Parameters used for dam-break test.

density	ρ	1000 kg/m^3
spatial step	dr	0.005 m
num. sound speed	c	120 m/s
gravitational constant	g	9.8 m/s^2
water column width	l_{wcw}	1 m
water column height	l_{wch}	2 m
box width	l_{bw}	4 m
box height	l_{bh}	3 m
kernel support radius	h	3 dr
Lennard-Jones radius	r_{wall}	0.95 dr
Lennard-Jones energy	E_{wall}	$10\,mgl_{\mathrm{wch}}$
time step	dt	$0.2\,h/c$

Besides being symplectic, Verlet scheme is also time reversible, which means that after changing the sign of velocities, the scheme returns back in time. The scheme thus preserves the time-reversal symmetry of the original system (1a), (1b), (1c). We can verify this by substituting

$$\boldsymbol{u}(t_m) \mapsto -\boldsymbol{u}(t_{m+1})$$
$$\boldsymbol{u}(t_{m+1}) \mapsto -\boldsymbol{u}(t_m)$$
$$\boldsymbol{r}(t_m) \mapsto \boldsymbol{r}(t_{m+1})$$
$$\boldsymbol{r}(t_{m+1}) \mapsto \boldsymbol{r}(t_m)$$

which produces the same set of equations

1. $-\boldsymbol{u}(t_{m+\frac{1}{2}}) = -\boldsymbol{u}(t_{m+1}) + \frac{1}{2}\delta t\, \boldsymbol{a}(\boldsymbol{r}(t_{m+1}))$
2. $\boldsymbol{r}(t_m) = \boldsymbol{r}(t_{m+1}) - \delta t\, \boldsymbol{u}(t_{m+\frac{1}{2}})$
3. $-\boldsymbol{u}(t_m) = -\boldsymbol{u}(t_{m+\frac{1}{2}}) + \frac{1}{2}\delta t\, \boldsymbol{a}(\boldsymbol{r}(t_m))$,

but in the reversed order. Unfortunately, the time reversibility fails in the floating-point arithmetic (FloPA), where addition is *not* associative. In particular, addition and subtraction of a value δx to a float x does not recover x. For instance

$$(1 + 0.5\epsilon) - 0.5\epsilon \overset{\text{float}}{=} 0.9999999999999999 \neq 1. \tag{8}$$

Although this error is usually very small, it tends to accumulate in the presence of non-linearities and destroys the reversibility in longer simulations.

There is, however, an easy solution suggested by [29] in the context of N-body simulations of Solar System. It converts vectors $\boldsymbol{r}(t_m)$, $\boldsymbol{u}(t_m)$, and $\boldsymbol{a}(t_m)$ to the fixed-point arithmetic (FixPA) just before they are used in the time-step evaluation. Since addition is associative in FixPA, this method allows for bit-precise time reversibility and long-time conservation of energy – see Fig. 3. Note that the intermediate computations of the sums in (5) and (1b) can be still performed in FloPA and only then converted to FixPA. However, due to the non-associativity of FloPA, summation of more than two elements in FloPA is order-dependent, and to ensure reversibility, the order of the summands must be chosen in a way that does not depend on the current state of the simulation (for instance, it is possible to perform summations always in the order of the particle indices).

In summary, symplectic WCSPH with the correct closed formula for density, the initial state correction, the conservative wall-fluid interaction, and the symplectic numerical Verlet scheme in the fixed-point arithmetic finally lead to globally-in-time reversible simulations. We have demonstrated the reversibility for instance on the dam break benchmark, and Appendix A contains the Gresho vortex benchmark. Although the simulation is symmetric, we can still observe the growth of entropy if we choose not to see all the details of the simulation, which is the matter of the following section (Fig. 4).

5. Dissipativity vs. decrease in mechanical energy

Can we judge dissipativity of an SPH simulation from observing decrease of mechanical energy or increase of internal (or compressional) energy? The purpose of this Section is to discuss the relation between dissipation and transformation of particular forms of energy.

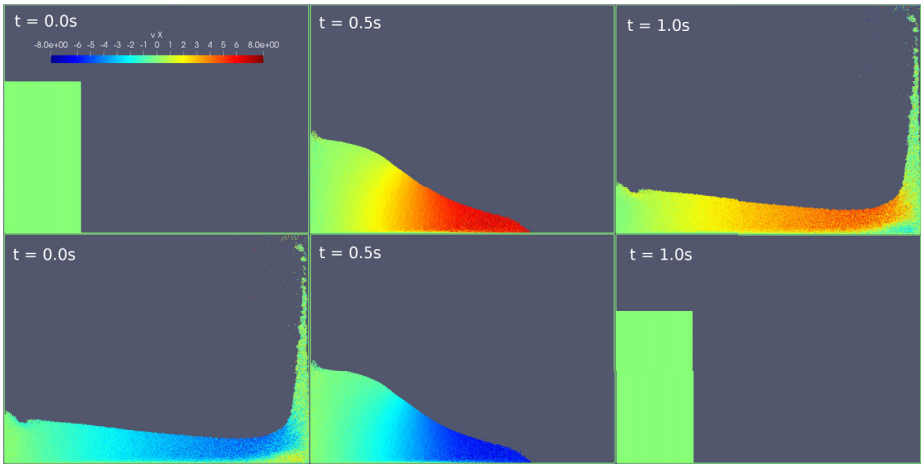
For instance the δ-SPH method adds a diffusive term to the continuity equation. The purpose of this term is stabilization of the scheme, and the term actually removes the noise from the pressure field and decreases the energy dissipation caused by the numerical scheme [17]. However, it is not straightforward to add the δ-SPH term into our symplectic scheme because our scheme does not contain the discretized continuity equation, to which that term would be added. Instead, we evaluate the density from the current particle positions (and from a correction based on their initial position). However, since our numerical scheme preserves the energy (up to a tiny constant that decreases with the time step), we do not need the δ-SPH term to decrease the dissipation of energy here, see Fig. 5. Moreover, since our aim is to obtain the global-in-time reversibility, we can not add the δ-SPH term here because it would violate the reversibility.

Nevertheless, the δ-SPH-like term might be added into the symplectic scheme in an indirect way. If another vector field is added to the state variables and coupled to the density as in mechanics of superfluids [33], it might be possible to keep the symplecticity of the SPH scheme. Algebraic dissipation in the equation for the extra vector field should then, after the field relaxes to a quasi-equilibrium value, restore a diffusion-like term in the density equation, as in [34].

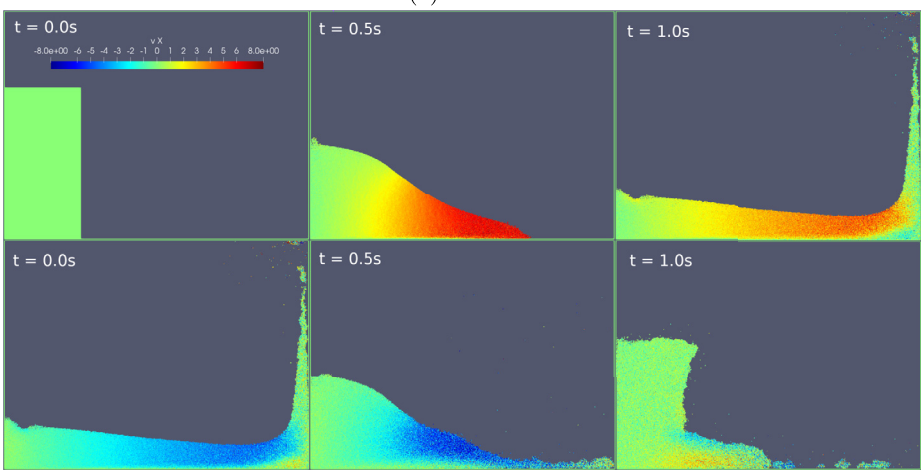
In paper [17], the sum of the kinetic and potential energies is called mechanical, and our internal energy is called compressional. The dissipation is calculated as the negative of the evolution of the mechanical energy in [17]. Therefore, we can compare the behavior of our kinetic, potential, and internal energies with the mechanical and compressional energies from [17]. Plots of the components of energy during the globally reversible dam-break simulation, namely the kinetic energy, gravitational (or potential) energy, the compressional energy, and the wall energy (from the wall-fluid interactions) are contained in Appendix C.

Since our scheme is symplectic, we preserve the underlying geometrical structure of the evolution equations (symplecticity), but the total energy is conserved only approximately. This is a feature of symplectic integrators that can not be circumvented because only the exact solutions preserve both the symplectic structure and energy [35]. However, the error in the total energy does not accumulate over time with symplectic integrators (it remains bounded by the time-step and order of the method), see [12] (Ch. IX.8). In our case, the numerical error in the total energy was less than $2 \cdot 10^{-4}$%. Moreover, the qualitative behavior (growth of entropy and global reversibility) is not affected by either raising or reducing the time-step, despite the numerical error in energy decreases with lower time steps. Therefore, the qualitative behavior is not caused by the numerical errors in the total energy.

While the total energy is approximately conserved in the dam-break simulation, the gravitational energy oscillates until it reaches the minimum when the particles occupy the bottom of the container. Similarly, the wall-energy oscillates during the simulation, but it is small in comparison with the total energy. The kinetic energy goes in a non-monotonous way from the initial zero (particles not moving initially) until it reaches its maximum. Similarly, the compressional (or internal) energy grows until it reaches its maximum. Energy mainly flows from the gravitational energy (which decreases in time) to the kinetic energy and to the internal (compressional) energy (approximately 3/4 to the kinetic and 1/4 to the internal).



(a) FixPA



(b) FloPA

Fig. 3. Results for dam break test with parameters from Table 1 using symplectic WCSPH. At $t = 1$ s, we reset the time counter and revert all velocities. In FixPA, the simulation is completely reversible and returns to the initial state. In FloPA, this fails due to round-off errors illustrated in (8).

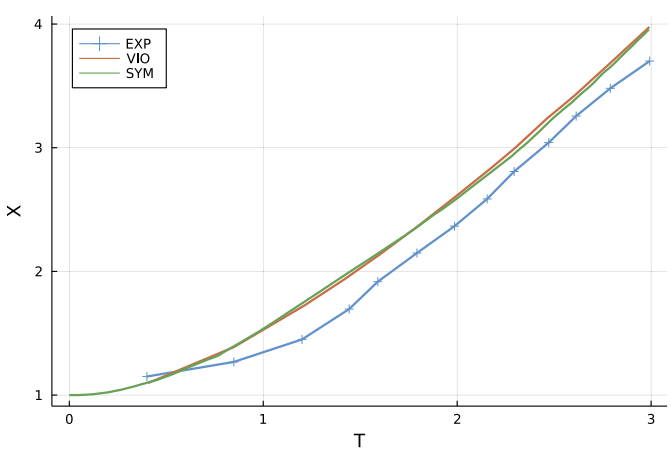


Fig. 4. Comparison of symplectic WCSPH (SYM) in dam-break simulation with experimental data [30] (EXP) and numerical result from Violeau's book [14] (VIO). Here, $T = t\sqrt{g/l_{\mathrm{wcv}}}$ is a dimensionless time and $X = l_{\mathrm{le}}/l_{\mathrm{wcv}}$, where l_{le} is the x-coordinate of the wave's leading edge.

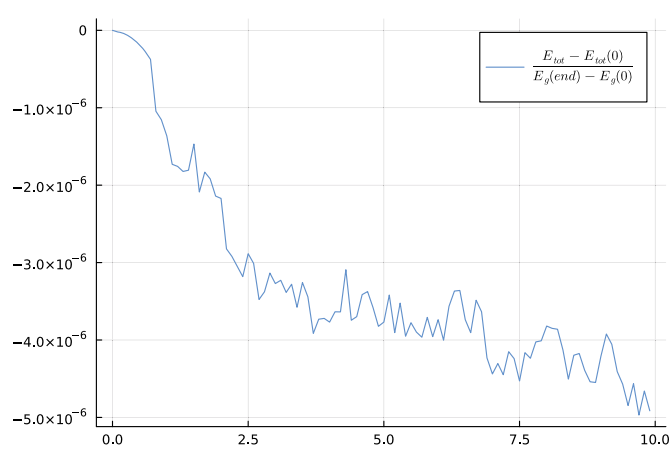


Fig. 5. Total energy (kinetic, potential, compressional, and wall) in the dam-break simulation. The initial value is subtracted and the remainder is divided by the total change of the potential energy during the simulation, as in [31,32]. The total energy is well preserved in our scheme. The total change is bounded in time and the bound decreases with the time step.

Although the overall mechanical energy (kinetic + potential) decreases, its drop can not be associated with dissipation in general. Instead, dissipation should be considered as the reduction of the total energy (including the compressional part). When this occurs, it also means that the simulated physical system is not isolated, since otherwise the total energy would be conserved. A typical SPH simulation is not isolated, as the total energy typically decreases due to either the integrator or due to artificial dissipation [17].

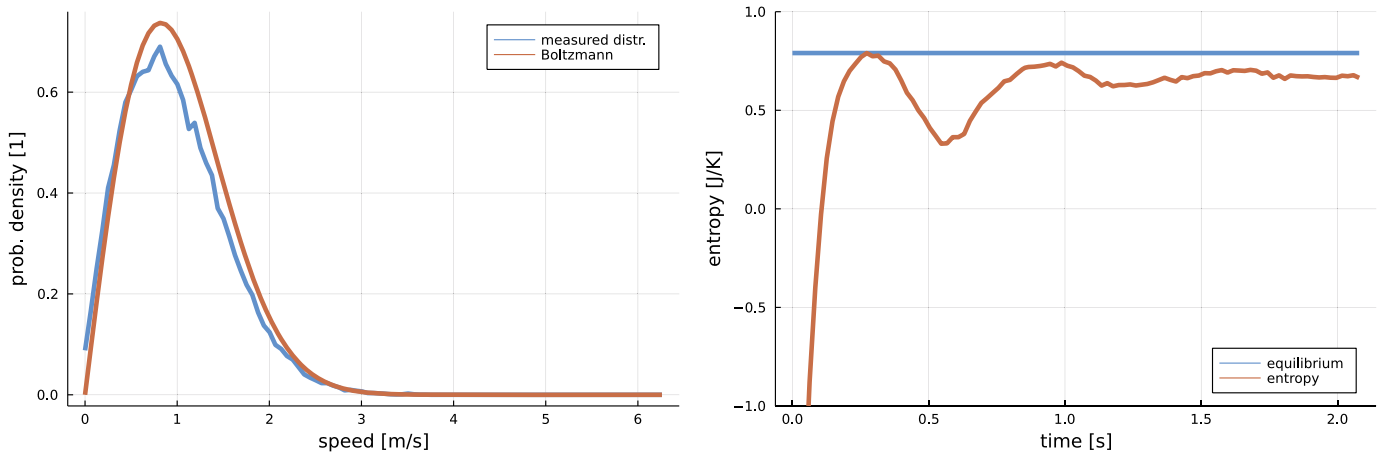


Fig. 6. *Left:* Histogram of the velocities of the particles at time $t = 2.0$ (blue) in comparison with the equilibrium Maxwell-Boltzmann distribution (orange) (19) with fitted temperature T. *Right:* Evolution of Boltzmann entropy (see Appendix B) in the reversible breaking-dam simulation (blue). Despite the reversibility of the dynamics, the entropy increases and approaches equilibrium entropy corresponding to the temperature of the measured Maxwell-Boltzmann distribution (blue line) from (21).

Let us now address the question whether symplectic integrators reduce the dissipation. First, symplectic integrators are suitable only for the reversible part of the differential equations, whereas the irreversible part (viscosity and diffusion) is typically not in the symplectic form [14]. Non-symplectic integrators, however, can cause irreversibility even in the reversible part of the equations. Although they can conserve the energy precisely, they do not preserve the phase-volume, which means that they affect the evolution of the distribution function of the particles [13]. Consequently, they affect the Liouville entropy of the system, which causes dissipation. Another source of irreversibility is the floating-point arithmetic, which actually spoils the reversibility and phase-volume preservation of the exact symplectic integrators, leading to similar source of irreversibility as the non-symplectic integrators. In practice, the dissipation caused by imperfect integration of the equations is often shadowed by the dissipation due to the viscous terms that are pertinent in situations that require some smoothing (for instance distorted boundary [15]). Nevertheless, if one wishes to minimize the dissipation due to the reversible terms in the SPH scheme, one should use the symplectic integrator, fixed-point arithmetic, and better also the other tweaks proposed in the current manuscript.

6. Emergence of the second law of thermodynamics

The second law of thermodynamics tells that the entropy of each isolated system grows until the system reaches the thermodynamic equilibrium. Where does this irreversible behavior come from when the underlying evolution equations for particle dynamics (here SPH) are reversible? In this section we illustrate that the emergence of the second law from completely reversible dynamics is caused by ignoring details of the dynamics, similarly as in [36–38].

The emergence of the second law is actually expected due to the result of Lanford [39] and following works [40], where a system of classical particles with a short-range potential is shown to obey the Boltzmann equation. The collision term in the Boltzmann equation then causes the growth of the Boltzmann entropy. Fig. 6 shows the growth of Boltzmann entropy in a reversible SPH breaking-dam simulation. But how to interpret the Lanford's mathematical result and the observed growth of Boltzmann entropy of the SPH particles from the physical point of view? The SPH particles obey reversible Hamiltonian dynamics. Another possibility to describe their motion is the Liouville equation, which expresses reversible evolution of the N-particle distribution function, $f_N(t, \mathbf{r}_1, \mathbf{p}_1, \dots, \mathbf{r}_N, \mathbf{p}_N)$. Indeed, Liouville equation follows from the particle mechanics, and, vice versa, if we set the distribution function to be the product of Dirac δ-distributions in the positions and momenta of all particles, we recover Hamilton canonical equations. Liouville equation also conserves the Liouville entropy

$$S^{\text{Liouville}}(f_N) = -\frac{k_B}{N!}\int d\mathbf{r}_1 \int d\mathbf{p}_1 \dots \int d\mathbf{r}_N \int d\mathbf{p}_N f_N \ln\left(h^{3N} f_N\right), \tag{9}$$

which thus remains constant. This is consistent with the reversibility of the underlying Hamiltonian particle mechanics.

Now we decide not to see all the positions and momenta of the individual particles, observing for instance only the probability distribution of momenta at given space and time. Such one-particle distribution function $f(t, \mathbf{r}, \mathbf{p})$ indeed does not contain the knowledge of positions and momenta of all particles and thus the entropy of the system expressed in terms of f must be higher than the Liouville entropy (functional of f_N). The former is the Boltzmann entropy and it is obtained by maximization of the Liouville entropy subject to the constraint that the one-particle distribution function is known [26]. Plugging the resulting f_N back into the Liouville entropy, we obtain the Boltzmann entropy

$$S^{\text{Boltzmann}}(f) = -k_B \int d\mathbf{r} d\mathbf{p} f (\ln\left(h^2 f\right) - 1) \tag{10}$$

see [26]. Because the Boltzmann entropy is the Liouville entropy evaluated at the point f_N where it is maximal (keeping the knowledge of f), the Boltzmann entropy is higher than the Liouville entropy. Although the Liouville entropy remains constant in the dynamics, the Boltzmann entropy can grow in the dynamics following the Liouville equation.

The irreversibility of the dynamics of the one-particle distribution function can be illustrated on the reversible breaking-dam simulation with fixed-point arithmetic. Section 4 contains details of the simulation, but let us also show the histogram of the particle velocity distribution in the middle of the simulation, just before the velocities are inverted. Fig. 6 shows the measured histogram of the velocities of the SPH particles in comparison with the equilibrium Maxwell-Boltzmann distribution function (19) obtained by

fitting the effective temperature.[7] Although all velocities were initially zero, their distribution approaches the equilibrium distribution. This approach towards equilibrium is reflected in the growth of Boltzmann entropy during the long-time reversible simulation.

In other words, when the particles are described by the Boltzmann equation, our knowledge is incomplete, since the positions and momenta of all the particles remain unknown, in contrast to the description by means of Hamilton canonical equations or Liouville equation. And this lack of knowledge makes the evolution of the one-particle distribution function f irreversible. The irreversibility is then explicitly expressed by the collision integral in the Boltzmann equation. The second law of thermodynamics thus emerges from completely reversible dynamics when our description is incomplete (not seeing all positions and momenta of the particles).

In summary, if we see all the positions and momenta in the SPH simulation, we can not see the second law of thermodynamics. Indeed, the simulation is reversible and the Liouville entropy remains constant. However, when we only focus on the one-particle distribution function, we can see the growth of Boltzmann entropy and thus irreversible behavior. The second law of thermodynamics emerges when we our knowledge about the precise state of the system is incomplete [38].

Interestingly, since our discrete system is deterministic, reversible and can exist in only finitely many states, it is recurrent [41]. In other words, if the system evolves long enough, it comes back to the initial state exactly (provided that the particles are not allowed to escape to infinity). However, since the phase space of the system contains thousands of SPH particles, it has enormous amount of states, which makes it highly unlikely to observe the recurrence theorem in practice.

7. Conclusion

In this paper we have turned the usual weakly compressible smoothed particle hydrodynamics (WCSPH) to a symplectic form by finding the correct closed formula for the mass density. Because the closed formula for density leads to different treatment of particles that are initially near the boundary, the method of initial state correction (ISC) was introduced, due to which all particles are then treated in the same way. This leads to stable SPH simulations in the presence of free surfaces without any further stabilization.

In order to get simulations that preserve the energy, we use a conservative fluid-wall interaction. A symplectic (Verlet) scheme, which is suitable for the symplectic WCSPH, implemented in the fixed-point arithmetic then leads to globally-in-time reversible SPH simulations. This is demonstrated on the dam-break benchmark, where inversion of velocities at a later stage of the simulation eventually leads the system back to the initial state. The simulations are available in a new Julia package *SmoothedParticles.jl* [20].

Despite the global reversibility of the simulations, we observe thermodynamic behavior when we do not use all the details of the simulation. The Boltzmann entropy, which depends only on the one-particle distribution function and not on positions of individual particles, grows in the dam-break simulation and approaches the equilibrium value. In other words, we observe the emergence of the second law of thermodynamics from purely reversible dynamics, caused by reduction of our knowledge about the system.

In future, we would like to extend the WCSPH framework to non-isothermal fluids and solids while keeping the Hamiltonianity of the equations and numerical schemes. In particular, we would like to apply our approach towards reversibility in solid mechanics, using for instance the Lagrangian kernels [9,10].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Our code, including the examples, is publicly available as the SmoothedParticles.jl Julia package.

Acknowledgements

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Appendix A. Gresho vortex benchmark

The Gresho vortex benchmark [42] prescribes a tangential velocity component of Eulerian fluid in polar coordinates as[8]

$$u_\theta^0(r) = \begin{cases} 5r & \text{for} \quad r < \frac{1}{5}, \\ 2 - 5r & \text{for} \quad \frac{1}{5} < r < \frac{2}{5}, \\ 0 & \text{for} \quad r > \frac{2}{5}. \end{cases}$$

The vortex is confined in a box $(-\frac{1}{2}, \frac{1}{2}) \times (-\frac{1}{2}, \frac{1}{2})$ with no-slip wall (though different variants can be found in literature). Theoretically, in the absence of viscosity, the vortex should be stationary, but obtaining this in a numerical scheme is challenging. In our numerical experiment, we simulate the flow in time-interval $[0, 1]$ and measure the error using a discretized (L^∞, L^2) Sobolev norm

$$e = \max_{t \in [0,1]} \sqrt{\frac{75}{4\pi} \sum_a V_a \left| \boldsymbol{u}_a - \boldsymbol{u}^0(\boldsymbol{r_a}) \right|^2}. \tag{11}$$

The meaning of the factor $\frac{75}{4\pi}$ is that the zero velocity field corresponds to error 1. In order to prevent formation of void space in the center of vortex, we had to include an anti-clump force term in the form

$$\boldsymbol{f} = \sum_b m_b \left(\frac{p_0}{\rho_a^2} + \frac{p_0}{\rho_b^2} \right) \tilde{w}'(r_{ab}) \boldsymbol{e}_{ab},$$

where $\tilde{w}$ is a smoothing kernel with support radius $\frac{\mathrm{dr}}{2}$. Additionally, we consider noise reduction by the Shepard filter

$$\tilde{\boldsymbol{u}}_a = \frac{1}{\gamma_a} \sum_b V_b \boldsymbol{u}_b w_{ab},$$

where

$$\gamma_a = \sum_b V_b w_{ab}.$$

[7] Note that the effective temperature is not the physical temperature of the system because it is calculated from the overall motion of the macroscopic SPH particles. To see this, consider the extreme case of only one SPH particle with weight one kilogram, where the effective temperature obtained from the motion of that macroscopic particle does not coincide with the physical temperature of that particle.

[8] Similar simulation of the Kepler vortex shows stability of the vortex even after 200 revolutions [43,44].

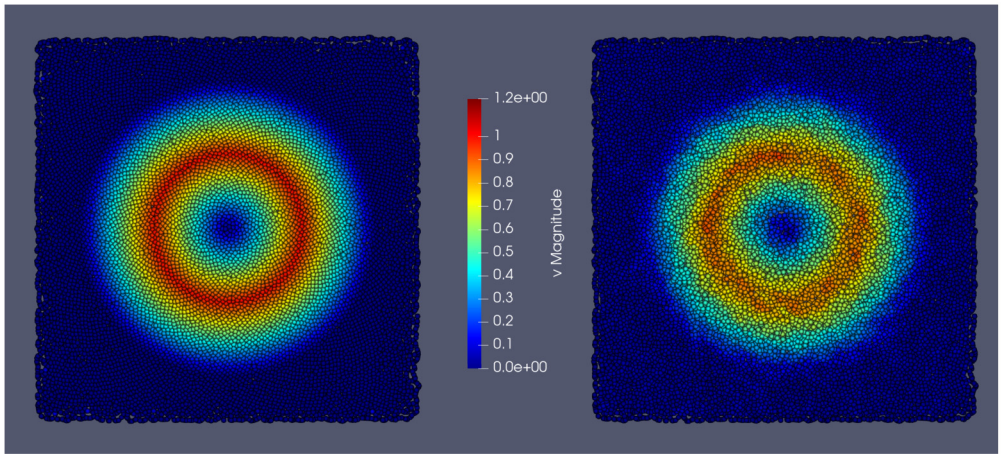

Fig. 7. Simulation result with Vogel spiral + ISC arrangement at $t = 0$ (left) and $t = 1$ (right).

Table 2
Error of Gresho vortex benchmark for different grid types (square, hexagonal, Vogel spiral, Vogel spiral + ISC) and with different types of noise filtering. The error is computed according to formula (11). It appears from data that passive (a posteriori) filter is better than active filtering (which dissipates energy). Compared to WENO method [42], numerical dissipation in symplectic WCSPH is quite strong.

	no filter	passive filter	active filter
square	28.16%	12.20%	53.07%
hexagonal	27.84%	11.77%	55.79%
Vogel	29.85%	13.31%	78.06%
Vogel + ISC	28.19%	11.26%	55.25%

Table 3
Parameters used for Gresho vortex simulation. This benchmark is dimensionless and with no gravity. Walls were implemented by two layers of dummy particles.

density	ρ	1
spatial step	dr	1e-2
num. sound speed	c	20
kernel support radius	h	$3\,dr$
box size	l	1
anti-clump pressure	p_0	10
time step	dt	$0.1h/c$
filter frequency	M	30

This filter can be applied either in post-processing (passive filter), or by setting $\boldsymbol{u}_a := \tilde{\boldsymbol{u}}_a$ every M time steps (active filter). Results are summarized in Table 2 and Fig. 7. Simulation parameters are listed in Table 3. Note that the pressure field is noisy because the noise is not suppressed by artificial dissipation [17]. However, the total energy is still conserved despite the noise. As the noise affects the quality of the solution, we opt-out for filtering it out in the post-processing rather than eliminating it in the calculation because the latter would violate our goals of global-in-time reversibility and energy conservation.

Appendix B. The Maxwell-Boltzmann entropy

In this section we derive the formula for Boltzmann entropy in terms of the Maxwell-Boltzmann distribution and, subsequently, to find its equilibrium value.

B.1. The reduced Boltzmann entropy in terms of the Maxwell-Boltzmann distribution

Let us start with the Boltzmann entropy in two dimensions expressed in terms of the one-particle distribution function $f(t, \mathbf{r}, \mathbf{p})$,

$$S^{(\text{Boltzmann})} = -k_B \int d\mathbf{r} \int d\mathbf{p} f \left(\ln\left(h^2 f\right) - 1\right), \tag{12}$$

where the position is constrained to a box with volume V and where h is the Planck constant. This formula can be obtained for instance by the principle of maximum entropy from the Liouville entropy, from where it also follows that the one-particle distribution function is normalized to the number of particles, see [26].

Assuming that the distribution function depends only on the magnitude of the momentum (isotropic dependence), the Boltzmann entropy can be rewritten as

$$S^{(\text{Boltzmann})} = k_B N \ln \frac{e}{h^2} - 2\pi k_B V \int_0^\infty dp\, p f(p) \ln f(p), \tag{13}$$

where $p = |\mathbf{p}|$ is the norm of the momentum. This dependence on the one-particle distribution function $f(p)$ has to be converted to a dependence on the Maxwell-Boltzmann distribution function $f_{MB}(v)$, which tells the probability that the norm of velocity of a particle $v = p/m$ is in the interval $(v, v + dv)$. This is done using the normalization,

$$1 = \frac{1}{N} \int d\mathbf{r} \int d\mathbf{p} f = \int_0^\infty dv\, \frac{2\pi m^2 V}{N} v f(v), \tag{14}$$

which leads to the expression of the two-dimensional Maxwell-Boltzmann distribution function in terms of the one-particle distribution,

$$f_{MB}(v) = \frac{2\pi m^2 V}{N} v f(v), \tag{15}$$

which is normalized to unity. Note that we assume that the one-particle distribution is homogeneous in space (the total volume being V) and isotropic in momentum. These assumptions are valid in the thermodynamic equilibrium, but they approximately hold also before the equilibrium is reached, and for the purpose of showing that the Boltzmann entropy grows in our simulations, the approximation is satisfactory. Finally, the Boltzmann entropy in terms of the Maxwell-Boltzmann distribution function becomes

$$S^{(\text{Boltzmann})} - k_B N \ln \frac{e}{h^2} = -k_B N \int_0^\infty dv\, f_{MB}(v) \ln\left(\frac{f_{MB}(v) N}{2\pi m^2 v}\right). \tag{16}$$

Since the number of particles is conserved in our simulation, we only evaluate the part of Boltzmann entropy that varies when f_{MB} changes,

$$S^{(\text{Boltzmann})}_{\text{reduced}} = \frac{S^{(\text{Boltzmann})} - k_B N \ln \frac{e}{h^2}}{k_B N} + \ln\left(\frac{N}{2\pi m^2}\right)$$
$$= -\int_0^\infty f_{MB}(v) \ln\left(\frac{f_{MB}(v)}{v}\right), \tag{17}$$

called reduced Boltzmann entropy. Note that the integral converges near the origin because the Maxwell-Boltzmann distribution typically has linear growth there.

In our simulations, this reduced Boltzmann entropy is numerically integrated using the approximate Maxwell-Boltzmann distribution function obtained from a histogram of the particle velocities at given time instant.

B.2. Equilibrium Boltzmann entropy

What is the final value of the reduced Boltzmann entropy when the system of particles reaches the thermodynamic equilibrium? This question can be answered in two steps. First, the equilibrium distribution function is calculated by the principle of maximum entropy (MaxEnt). Second, the equilibrium distribution function is plugged into the formula for the reduced Boltzmann entropy. Note, however, that the MaxEnt step depends on the energy (Hamiltonian) of the system and it depends on the complexity of the Hamiltonian whether the calculation can proceed purely analytically (without numerical solutions). Therefore, we use the simplest Hamiltonian approximating the true Hamiltonian of our SPH particles, consisting only of the kinetic energy of the particles. Maximization of the Boltzmann entropy subject to the constraints given by the total energy and total number of particles leads to the equilibrium one-particle distribution function

$$f_{\text{equilibrium}} = \frac{1}{h^2} e^{-\frac{N^*}{k_B}} e^{-\frac{E^*}{k_B}\frac{\mathbf{p}^2}{2m}}, \tag{18}$$

where N^* and E^* are the Lagrange multipliers corresponding to the two constraints (number of particles and total energy). From the normalization it follows that $N/V = \exp(-N^*/k_B)/h^2$, while the other Lagrange multiplier can be interpreted as the inverse temperature, $E^* = 1/T$.

The equilibrium Maxwell-Boltzmann distribution function then becomes

$$f_{MB,\text{equilibrium}}(T, v) = \frac{m}{k_B T} v e^{-\frac{1}{2}mv^2}. \tag{19}$$

As time proceeds in our simulations, the histogram of particle velocities approaches the equilibrium Maxwell-Boltzmann distribution.

When this equilibrium distribution function is plugged back into the Boltzmann entropy, the reduced Boltzmann entropy becomes

$$S^{(\text{Boltzmann})}_{(\text{reduced})}(T) = 1 + \ln\left(\frac{k_B T}{m}\right). \tag{20}$$

This equilibrium entropy, which depends on T, can be calculated once the temperature is obtained by fitting the histogram of particle velocities to the equilibrium Maxwell-Boltzmann distribution function (19). This makes sense, however, only at later stages in the simulations, when the histogram approaches the equilibrium distribution.

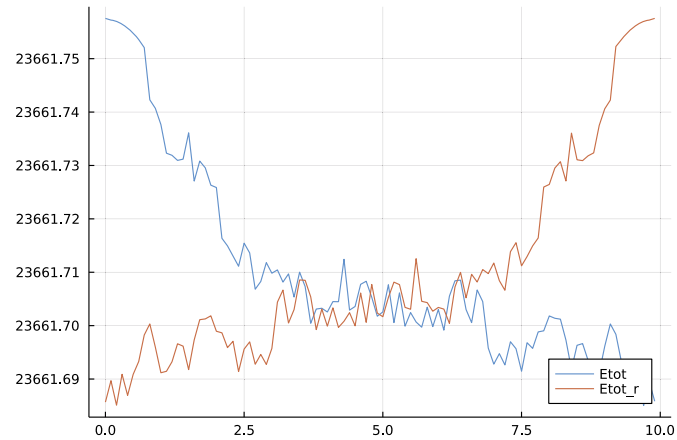

Fig. 8. Evolution of the total energy in the dam-break simulation (forward and reversed simulations). The energy oscillates within $2 \cdot 10^{-4}$% of the initial (exact) value.

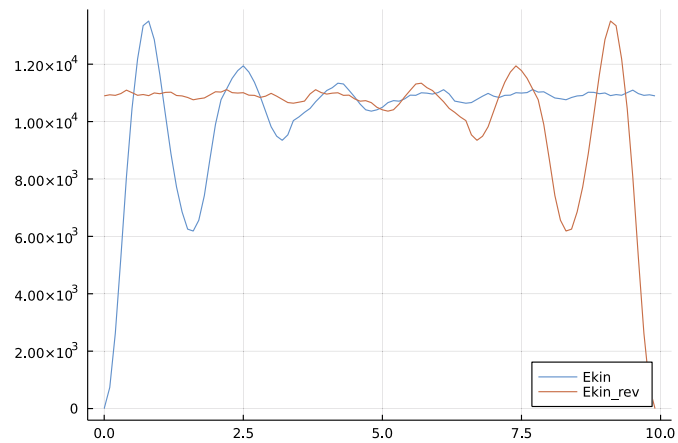

Fig. 9. Kinetic energy in the dam-break simulation (forward and reversed simulations). The energy starts from zero and then evolves in a non-monotonous way towards its maximum when the particles occupy the bottom of the container.

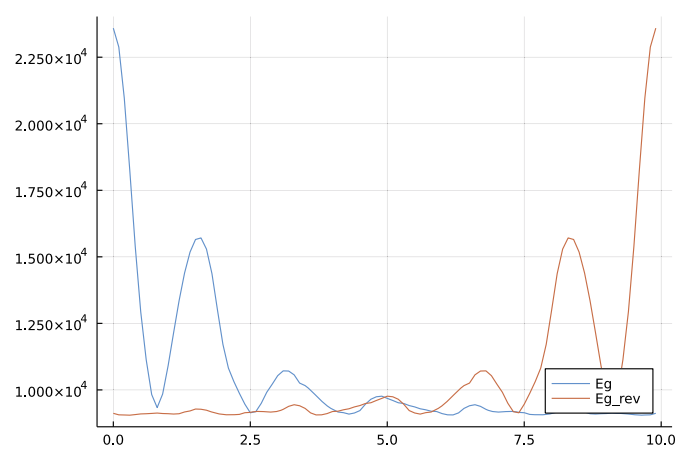

Fig. 10. Gravitational (potential) energy in the dam-break simulation (forward and reversed simulations). The energy decreases as the particles tend to occupy the bottom of the container.

Instead of using the equilibrium temperature, which makes sense only in later stages of the simulations, we can express the equilibrium entropy in terms of energy, which can be measured anytime. The total kinetic energy of the two-dimensional system of particles is equal to $E = Nk_B T$, which makes it possible to write the equilibrium entropy in terms of the energy,

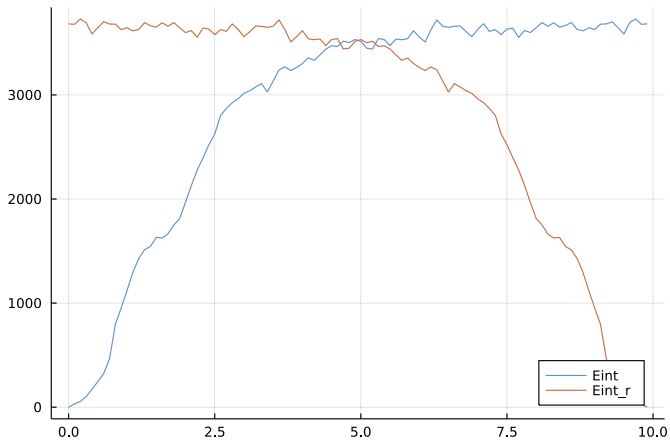

Fig. 11. Internal (or compressional) energy in the dam-break simulation (forward and reversed simulations). The energy grows from the initial condition to its maximum when the particles occupy the bottom of the container.

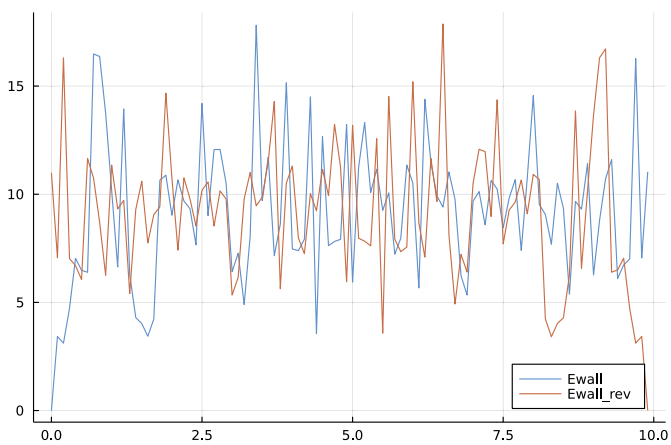

Fig. 12. The wall-fluid interaction energy in the dam-break simulation (forward and reversed simulations). Once the fluid starts interacting with the wall, the wall energy oscillates around an average value.

$$S^{(\text{Boltzmann})}_{(\text{reduced})}(E) = 1 + \ln\left(\frac{E}{Nm}\right). \tag{21}$$

This value of equilibrium entropy is close to the Boltzmann entropy obtained directly from the approximated Maxwell-Boltzmann distribution, while the entropy based on the equilibrium temperature is only approached at later stages of the simulations.

Appendix C. Energy budget in the dam-break simulation

This Section contains the plots of various parts of the energy (gravitational, kinetic, internal, wall) and the total energy within the reversible dam-break simulation. The time-span ranges from the initial condition to the point where the particle velocities were inverted. The plots contain both the forward simulation and the backward simulation (denoted by subscript rev). See Figs. 8–12.

References

[1] R. Gingold, J. Monaghan, Mon. Not. R. Astron. Soc. 181 (3) (1977) 375–389.
[2] M. Antuono, A. Colagrossi, S. Marrone, D. Molteni, Comput. Phys. Commun. 181 (3) (2010) 532–549.
[3] C. Zhang, X.Y. Hu, N.A. Adams, J. Comput. Phys. 335 (2017) 605–620.
[4] D.D. Meringolo, S. Marrone, A. Colagrossi, Y. Liu, Comput. Fluids 179 (2019) 334–355.
[5] M. Svärd, Phys. A, Stat. Mech. Appl. 506 (2018) 350–375.
[6] M. Pavelka, V. Klika, M. Grmela, Phys. D: Nonlinear Phenom. 399 (2019) 193–210.
[7] Z.-F. Meng, A.-M. Zhang, P.-P. Wang, F.-R. Ming, Comput. Methods Appl. Mech. Eng. 386 (2021) 114082.
[8] J. Swegle, D. Hicks, S. Attaway, J. Comput. Phys. 116 (1) (1995) 123–134.
[9] T. Belytschko, Y. Guo, W. Kam Liu, S. Ping Xiao, Int. J. Numer. Methods Eng. 48 (9) (2000) 1359–1400.
[10] T. Rabczuk, T. Belytschko, Int. J. Numer. Methods Eng. 61 (13) (2004) 2316–2343.
[11] T. Rabczuk, T. Belytschko, Comput. Methods Appl. Mech. Eng. 196 (29) (2007) 2777–2799.
[12] E. Hairer, C. Lubich, G. Wanner, Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations, Springer Series in Computational Mathematics, Springer, Berlin Heidelberg, 2013.
[13] W. Hoover, T. Pierce, C. Hoover, J. Shugart, C. Stein, A. Edwards, Comput. Math. Appl. 28 (10) (1994) 155–174.
[14] D. Violeau, Fluid Mechanics and the SPH Method: Theory and Applications, Oxford University Press, Oxford, UK, 2012.
[15] A. Colagrossi, M. Landrini, J. Comput. Phys. 191 (2) (2003) 448–475.
[16] J.J. Monaghan, A. Rafiee, Int. J. Numer. Methods Fluids 71 (5) (2013) 537–561.
[17] M. Antuono, S. Marrone, A. Colagrossi, B. Bouscasse, Comput. Methods Appl. Mech. Eng. 289 (2015) 209–226.
[18] L. Landau, E. Lifshitz, Mechanics, Butterworth-Heinemann, 1976.
[19] L. Landau, E. Lifschitz, Statistical Physics. No. pt. 1 in Course of Theoretical Physics, Pergamon Press, 1969.
[20] O. Kincl, M. Pavelka, Smoothedparticles.jl, 2021.
[21] J. Bezanson, A. Edelman, S. Karpinski, V.B. Shah, SIAM Rev. 59 (1) (2017) 65–98.
[22] J. Monaghan, Rep. Prog. Phys. 68 (07 2005) 1703.
[23] M. Kroeger, M. Huetter, Comput. Phys. Commun. 181 (2010) 2149–2157.
[24] L. Verlet, Phys. Rev. 159 (Jul 1967) 98–103.
[25] J.J. Monaghan, J. Comput. Phys. 110 (2) (1994) 399–406.
[26] M. Pavelka, V. Klika, M. Grmela, Multiscale Thermo-Dynamics, de Gruyter, Berlin, 2018.
[27] B. Leimkuhler, S. Reich, C.U. Press, Simulating Hamiltonian Dynamics, Cambridge Monographs on Applied and Computational Mathematics, Cambridge University Press, 2004.
[28] J. Candy, W. Rozmus, J. Comput. Phys. 92 (1) (1991) 230–256.
[29] H. Rein, D. Tamayo, Mon. Not. R. Astron. Soc. 473 (09 2017) 3351–3357.
[30] S. Koshizuka, Y. Oka, Nucl. Sci. Eng. 123 (3) (1996) 421–434.
[31] S. Marrone, M. Antuono, A. Colagrossi, G. Colicchio, D. Le Touzé, G. Graziani, Comput. Methods Appl. Mech. Eng. 200 (13) (2011) 1526–1542.
[32] S. Marrone, A. Colagrossi, A. Di Mascio, D. Le Touzé, Phys. Rev. E 93 (May 2016) 053113.
[33] M. Sýkora, M. Pavelka, M. La Mantia, D. Jou, M. Grmela, Phys. Fluids 33 (127124) (2021).
[34] I. Peshkov, M. Pavelka, E. Romenski, M. Grmela, Contin. Mech. Thermodyn. 30 (6) (2018) 1343–1378.
[35] G. Zhong, J.E. Marsden, Phys. Lett. A 133 (3) (1988) 134–139.
[36] M. Kac, Bull. Cl. Sci., Acad. R. Belg. 5 (42) (1956) 356–361.
[37] P. Ehrenfest, T. Ehrenfest, The Conceptual Foundations of the Statistical Approach in Mechanics, Dover Books on Physics, Dover Publications, 1990.
[38] M. Pavelka, V. Klika, M. Grmela, J. Stat. Phys. 181 (1) (2020) 19–52.
[39] O.E. Lanford, in: J. Moser (Ed.), Dynamical Systems, Theory and Applications, in: Lecture Notes in Physics, vol. 38, Springer–Verlag, Berlin, 1975, pp. 1–111.
[40] M. Pulvirenti, C. Saffirio, S. Simonella, Rev. Math. Phys. 26 (02) (2014) 1450001.
[41] D. Wallace, J. Math. Phys. 56 (2) (2015) 022105.
[42] H. Grimm-Strele, F. Kupka, H. Muthsam, Comput. Phys. Commun. 185 (3) (2014) 764–776.
[43] A. Cartwright, D. Stamatellos, A.P. Whitworth, Mon. Not. R. Astron. Soc. 395 (2009) 2373–2380.
[44] L. Cullen, W. Dehnen, Mon. Not. R. Astron. Soc. 408 (07 2010) 669–683.